

Peptide identification using Mascot

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Practical 1

- a) Using the website uniprot.org, look for the FASTA file of insulin in dogs, and copy the text file.
- b) Perform the in silico digestion, using trypsin, and give the resulting peptides (without considering missed cleavages).
- b) for each resulting peptide with more than five amino acids (there should be six), and using

<http://db.systemsbiology.net:8080/proteomicsToolkit/FragIonServlet.html>

calculate the monoisotopic (M+H)⁺ mass of the peptide.

Practical 2

Using the Mascot tool (<http://www.matrixscience.com>) for peptide identification, determine which peptides are associated with the following fragmentation spectra (the corresponding zoom scan is indicated in small print). To do this, manually create an .mgf file with the approximate intensities observed. Use the non-redundant NCBI database (NCBInr) for all species, allowing at least one missed cleavage.

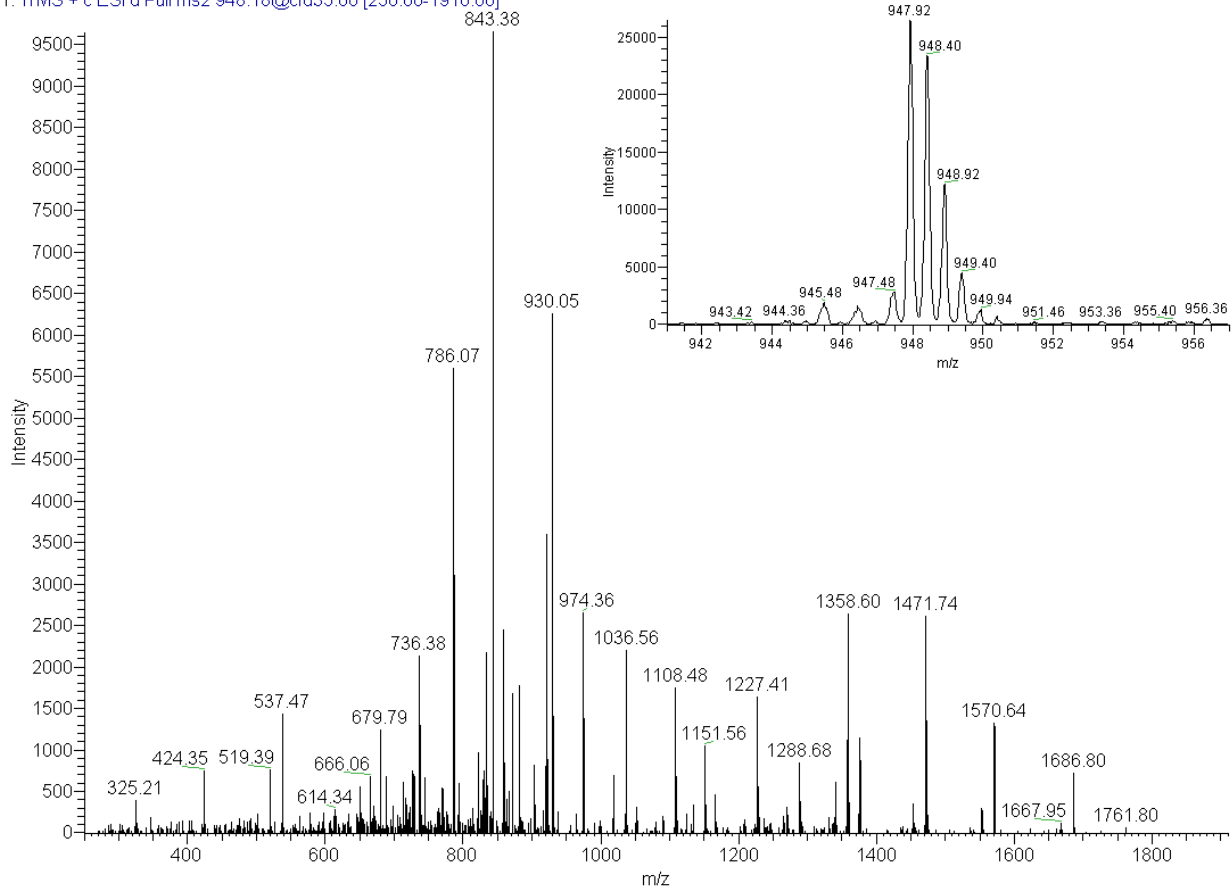
Note that:

- The spectra are from a SILAC experiment (lysines and arginines are labelled with ^{13}C instead of ^{12}C , increasing the total mass of these amino acids by +6 Da).
- We do not know whether the methionines will be oxidised or not.
- The sample has been treated with iodoacetamide.
- The peptides have been digested with trypsin.

a)

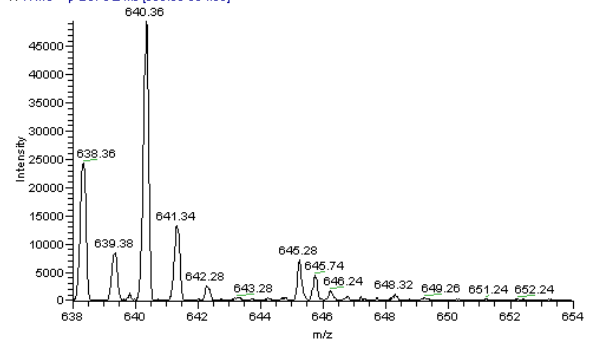
010609_SILAC_aliPrü_Sach#2641 RT: 58.96 AV: 1 NL: 9.65E3
T: ITMS + c ESI d Full ms2 948.18@cid35.00 [250.00-1910.00]

010609_SILAC_aliPrü_Sach#2640 RT: 58.94 AV: 1 NL: 2.65E4
T: ITMS + p ESI d Z ms [941.00-957.00]

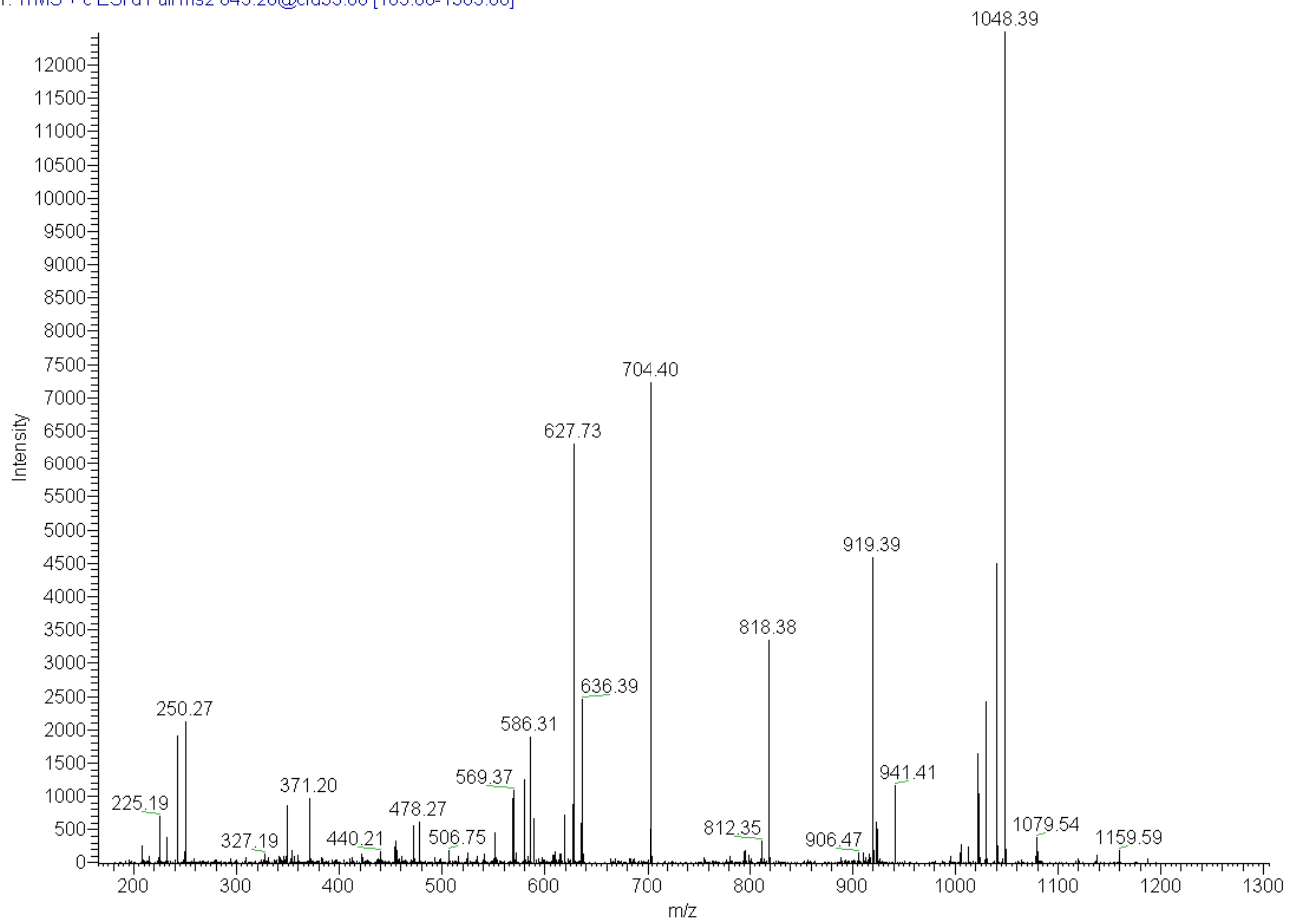


b)

010609_SILAC_alicPru_Sach #2739 RT: 60.50 AV: 1 NL: 4.96E4
T: ITMS + p ESI d Z ms [638.00-654.00]



010609_SILAC_alicPru_Sach #2740 RT: 60.52 AV: 1 NL: 1.25E4
T: ITMS + c ESI d Full ms2 645.26@cid35.00 [165.00-1305.00]



c)

010609_SILAC_aliPrü_Sach #2879 RT: 62.69 AV: 1 NL: 6.39E3
T: ITMS + c ESI d Full ms2 681.78@cid35.00 [175.00-2000.00]

010609_SILAC_aliPrü_Sach #2878 RT: 62.67 AV: 1 NL: 7.41E3
T: ITMS + p ESI d Z ms [674.00-690.00]

